SHASHAANK THUMMA

DATA ENGINEER & BI DEVELOPER

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PROFILE SUMMARY:

- Over 8 years of experience as a Data Engineer with expertise in designing, developing, and optimizing scalable data pipelines, ETL workflows, and cloud-based data solutions.
- Proficient in big data technologies including Apache Spark (RDD, Spark SQL, Spark Streaming), Hadoop (HDFS, MapReduce), Hive, Kafka, and Apache Flink for efficient data processing and real-time analytics.
- Skilled in programming languages such as Python (Pandas, NumPy), SQL (T-SQL, PL/SQL), Scala, and Java for building robust data engineering solutions.
- Extensive experience with cloud platforms like AWS (S3, Redshift, Lambda), Azure (Data Factory, Databricks, Synapse Analytics), and GCP (Big Query); adept at leveraging serverless cloud solutions such as AWS Lambda and Azure Functions for event-driven architectures.
- Expertise in data warehousing tools such as Snowflake, Redshift, Azure Synapse Analytics, Google BigLake, and Delta Lake for designing star schema models and optimizing query performance.
- Hands-on experience with ETL & data integration tools, including Apache Airflow, AWS Glue, Informatica PowerCenter, dbt (Data Build Tool), and Azure Data Factory (ADF) to automate workflows and ensure efficient data integration.
- Adept at building real-time streaming pipelines using Kafka integrated with Spark Streaming to process event-driven data.
- Proficient in creating interactive dashboards using BI tools like Power BI (DAX Expressions, Dataflows), Tableau (LOD Expressions), and Looker to enable actionable business insights.
- Strong knowledge of CI/CD pipelines using Jenkins, Docker, GitHub Actions, Kubernetes, and Terraform for automated deployment of workflows.
- Skilled in advanced SQL query optimization, Hive partitioning/bucketing, PySpark transformations, and PostgreSQL for enhanced performance of data processing tasks.
- Proficient in handling both structured and unstructured data with NoSQL databases like MongoDB and Cassandra.
- Experienced in migrating on-premises databases to cloud platforms like AWS Redshift and Azure Synapse Analytics to improve scalability and reduce costs.
- Hands-on with data lake architectures for distributed storage and processing using Azure Data Lake Service (Gen1/Gen2) and Hadoop clusters.
- Demonstrated success in implementing data governance practices, including role-based access control (RBAC) and validation rules to ensure high-quality data processing.
- Skilled in leveraging emerging technologies like Delta Lakehouse architecture to deliver impactful solutions that drive business value.
- Proven ability to optimize ETL processes using Python scripts and PySpark transformations to enhance performance by reducing job runtimes significantly.
- Experience with robust monitoring mechanisms using Apache Airflow DAGs for scheduling ETL workflows efficiently.

TECHNICAL SKILLS:

Languages	Python (Pandas, NumPy), SQL (T-SQL, PL/SQL), Scala, Java
BI Tools	Power BI (DAX, Dataflows), Tableau, Looker
ETL & Data Integration Tools	Apache Airflow, AWS Glue, Informatica PowerCenter, dbt (Data Build Tool), Azure Data Factory (ADF)
Cloud Platform	AWS (S3, Redshift, Lambda, Snowflake Integration), Azure (ADF, Databricks, Synapse), GCP (Big Query, Snowflake Intergration); Serverless Cloud Solutions (Azure Functions, AWS Lambda)
Data Warehousing	Snowflake (RBAC, Data Sharing, Performance Tuning, API Integration), Redshift, Azure Synapse Analytics; Google Big Lake and Delta Lake
Big Data Technologies	Apache Spark (RDDs, Spark SQL), Hadoop (HDFS, MapReduce), Hive, Databricks Kafka, Apache Flink; Spark Streaming
CI/CD & DevOps	Jenkins, GitHub Actions, Docker, Kubernetes, Terraform, CI/CD for Snowflake & Python-based ETL
Databases	Oracle, SQL Server, Azure SQL Database, Excel, MongoDB, Cassandra, PostgreSQL
API Development & Integration	Spring Boot, FastAPI, REST APIs for Data Integration

EXPERIENCE:

DATA ENGINEER — DEC- 2023 TO TILL DATE

CLIENT: NRG ENERGY, CHARLOTTE, NC

RESPONSIBILITIES:

- Designed and implemented scalable ETL pipelines using Python and Apache Airflow to automate data ingestion from diverse sources into Snowflake.
- Developed Spark applications in Scala to process multi-terabyte datasets, optimizing performance and reducing processing time by 30%.
- Migrated on-premises Oracle databases to AWS Redshift using AWS Glue and S3, improving query performance and reducing storage costs.
- Built real-time streaming pipelines using Kafka integrated with Spark Streaming to process event-driven data, enabling near real-time analytics.
- Created interactive dashboards in Tableau connected to Snowflake to provide actionable insights for business stakeholders, increasing reporting efficiency by 25%.
- Implemented CI/CD pipelines using Jenkins for automated deployment of ETL workflows across development, testing, and production environments.
- Enhanced ETL architecture by integrating data quality checks using Informatica PowerCenter and dbt transformations, ensuring high data accuracy.
- Conducted root cause analysis on failed workflows, implementing automated alert systems to minimize downtime during critical operations.
- Collaborated with data science teams to deliver pre-processed datasets for machine learning models, improving model training efficiency.

Environment: Apache Spark, Snowflake, AWS Glue, Kafka, Docker, Tableau, CI/CD, AWS Lambda

DATA ENGINNER MAR- 2022 TO NOV- 2023

CLIENT: MAYO CLINIC, ROCHESTER MN

RESPONSIBILITIES:

- Developed Azure Data Factory pipelines to orchestrate seamless data movement between on-premises systems and Azure Data Lake Storage.
- Wrote PySpark scripts to transform semi-structured JSON files into structured datasets for analytics use cases, reducing manual processing time by 40%.
- Designed Snowflake data models optimized for reporting requirements, enabling faster insights generation through Power BI dashboards.
- Migrated legacy ETL processes from Hadoop MapReduce to Apache Spark for improved processing efficiency, reducing job runtimes by 50%.
- Automated daily batch jobs using Python integrated with Linux shell scripting, ensuring timely delivery of critical business reports.
- Built RESTful APIs using Spring Boot for dynamic data retrieval from cloud-based storage systems, enabling integration with external applications.
- Conducted performance tuning of SQL queries and Spark jobs to optimize resource utilization on Azure Databricks clusters.
- Partnered with DevOps teams to implement containerized deployments using Docker for consistent workflow execution across environments.

Environment: Azure Data Factory (ADF), Databricks, Power BI (DAX Expressions), Snowflake, GCP BigQuery

DATA ENGINEER AUG- 2018 TO MAY 2021

CLIENT: VALUEFY SOLUTIONS, INDIA

RESPONSIBILITIES:

- Collaborated with business analysts to gather requirements and design star schema models for a centralized data warehouse, improving reporting accuracy.
- Developed Hive queries with partitioning and bucketing for efficient query execution on Hadoop clusters, reducing query latency by 20%.
- Built RESTful APIs using Spring Boot for dynamic data retrieval from cloud-based storage systems, enabling real-time reporting capabilities.
- Automated daily batch processing jobs using Python scripts integrated with Linux shell scripting, ensuring timely availability of processed data.
- Assisted in migrating on-premises Hadoop workloads to AWS EMR clusters, reducing infrastructure costs while maintaining scalability.
- Conducted detailed data profiling and implemented validation rules to ensure accuracy in ingested datasets before loading into Hive tables.
- Supported the development of Spark jobs for distributed processing of large datasets, contributing to faster ETL pipelines.

Environment: Hadoop (HDFS), Hive, AWS EMR, Spring Boot, Tableau

DATA ANALYST — NOV 2015 TO JUN-2018

CLIENT: VALUELABS, INDIA

RESPONSIBILITIES:

- Analyzed large datasets using Python (Pandas) and SQL to uncover trends and deliver actionable insights that improved operational decision-making by 15%.
- Developed interactive dashboards in Tableau and Power BI (Dataflows) to visualize key performance indicators (KPIs), increasing stakeholder engagement in analytics reviews.
- Conducted detailed data profiling and quality checks across financial reports, ensuring accuracy and compliance with regulatory standards.
- Automated reporting processes by creating Python scripts integrated with APIs for real-time updates on metrics, reducing manual effort by 30%.
- Collaborated with cross-functional teams to define reporting requirements and create ad-hoc reports tailored to business needs.
- Designed SQL queries for complex joins across multiple databases, enabling seamless access to consolidated datasets for analytics purposes.

Environment: Tableau, Power BI (DAX Expressions), Python (Pandas), SQL Server